Tut 1.

Ex. 8.1

8.6. $\displaystyle\int \frac{1}{x-\sqrt{x}}\,dx = \int \frac{1}{\sqrt{x}(\sqrt{x}-1)}\,dx = \int \frac{2\,du}{u-1} = ?$

$$\frac{1}{(x-\sqrt{x})}\cdot\frac{x+\sqrt{x}}{(x+\sqrt{x})}$$

$$= \frac{x+\sqrt{x}}{x^2-x}$$

$$= \frac{x+\sqrt{x}}{x(x-1)}$$

$$= \frac{1}{x-1} + \underbrace{\frac{\sqrt{x}}{x(x-1)}}_{\frac{1}{\sqrt{x}(x-1)}}$$

$$\frac{1}{x-\sqrt{x}} = \frac{1}{\sqrt{x}(\sqrt{x}-1)}$$

$$u = \sqrt{x} \Rightarrow du = \frac{1}{2\sqrt{x}}\,dx$$

$$\int \frac{1}{\sqrt{x}(x-1)}\,dx = \int \frac{2\,du}{u^2-1}$$

$$u^2-1 = (u+1)(u-1)$$

Q18 $\displaystyle\int \frac{2^{\sqrt{y}}}{2\sqrt{y}}\,dy = \int 2^u\,du = \frac{2^u}{\ln 2} + C$

$u = \sqrt{y} \Rightarrow du = \frac{1}{2\sqrt{y}}\,dy$ $\qquad = \dfrac{2^{\sqrt{y}}}{\ln 2} + C$

$\boxed{x^{\frac{n}{2}}}$ → Try substitution $\boxed{x^{\frac{n+2}{2}}}$

↳ appear in the derivative of $x^{3/2}$

$\boxed{x^{\frac{1}{2}}}$

Q40 $\displaystyle\int \frac{\overset{x^{\frac{1}{2}}}{\overset{\shortparallel}{\boxed{\sqrt{x}}}}}{1+x^3}\,dx = \frac{2}{3}\int \frac{du}{1+u^2} = \frac{2}{3}\arctan u + C$

$= \frac{2}{3}\arctan x^{\frac{3}{2}} + C$

Hint: $u = x^{3/2} \Rightarrow du = \frac{3}{2}x^{\frac{1}{2}}\,dx = \frac{3}{2}\boxed{\sqrt{x}\,dx}$

$\displaystyle\int \frac{x^{\frac{3}{2}}}{1+x^5}\,dx \qquad \int \frac{x^{3/2}}{\sqrt{1-x^5}}\,dx$

$$\int \frac{x^{\frac{3}{2}}}{1+x^5}\,dx$$

$$\int x^{3/2} f(x^5)\,dx = \frac{2}{5}\int f(u^2)\,du$$

$$u = x^{5/2} \Rightarrow du = \frac{5}{2}x^{3/2}\,dx$$

Q26

$$\int \frac{6}{\sqrt{y}(1+y)}\,dy = 12\arctan u + C = 12\arctan\sqrt{y} + C$$

$$\frac{1}{\sqrt{y}} f(y) \rightarrow 2f(u^2)$$

$$u = \sqrt{y}$$

Q38

$$\int \frac{d\theta}{\cos\theta - 1} = \int \frac{d\theta}{\cos\theta - 1} \cdot \frac{\cos\theta + 1}{\cos\theta + 1}$$

$$= \int \frac{\cos\theta + 1}{\cos^2\theta - 1}\, d\theta$$

$\sin^2\theta + \cos^2\theta = 1$

$\cos^2\theta - 1 = \boxed{-\sin^2\theta}$

① Mult. by conjugate

② Half Angle Formula

$$= \int \frac{\cos\theta + 1}{-\sin^2\theta}\, d\theta$$

$$= -\int \frac{\cos\theta}{\sin^2\theta}\, d\theta - \int \frac{1}{\sin^2\theta}\, d\theta$$

$$\int \frac{1}{\sin^2\theta}\, d\theta = \int \csc^2\theta\, d\theta = \cancel{-\ln|\ \ +\ \ |}$$

$$-\cot\theta + C$$

$$\int \frac{\cos\theta}{\sin^2\theta}\, d\theta = \int \frac{\cos\theta}{\sin\theta} \cdot \frac{1}{\sin\theta}\, d\theta = \int \cot\theta \csc\theta\, d\theta$$

$$= -\csc\theta + C$$

Q24 $\int (\sec t + \cot t)^2 \, dt$

$$= \int \sec^2 t \, dt + 2\int \sec t \cot t \, dt + \int \cot^2 t \, dt$$

$$\int \sec^2 t \, dt = \tan t + C$$

$$\int \sec t \cot t \, dt = \int \frac{1}{\cancel{\cos t}} \cdot \frac{\cancel{\cos t}}{\sin t} \, dt$$

$$= \int \frac{1}{\sin t} \, dt$$

$$= \int \csc t \, dt$$

$$= -\ln|\csc t + \cot t| + C$$

$$\int \cot^2 t \, dt = \int \csc^2 t - 1 \, dt = \underline{\underline{-\cot t - t}} + C$$

$s^2 + c^2 = 1$

$1 + \cot^2\theta = \csc^2\theta$

$\cot^2\theta = \underline{\csc^2\theta - 1}$

<u>Q8</u> $$\int \frac{2^{\ln z^3}}{16z} dz = \int \frac{2^u}{16 \cdot 3} du$$

$u = \ln z^3$

$= 3 \ln z$

$du = \frac{3}{1} \cdot \frac{1}{z} dz$

$$= \frac{1}{48} \int 2^u \, du$$

$$= \frac{2^u}{48 \ln 2} + C$$

$$= \frac{2^{3\ln z}}{48 \ln 2} + C$$

Q9

$$\int \frac{dz}{e^z+e^{-z}} \cdot \frac{e^z}{e^z} = \int \frac{e^z}{e^{2z}+1}\,dz$$

$$= \int \frac{e^z}{(e^z)^2+1}\,dz$$

$u = e^z$

$du = e^z\,dz$

$$= \int \frac{1}{u^2+1}\,du$$

$$= \arctan u + C$$

$$= \arctan e^z + C$$

$u = e^z + e^{-z}$

$du = e^z - e^{-z}\,dz$

$$\frac{1}{[f(x)]^2+1}$$

$$dz = \frac{1}{e^z - e^{-z}}\,du$$

$$\int \frac{1}{(e^z+e^{-z})(e^z-e^{-z})}\,du$$

$\hookrightarrow e^{2z} - e^{-2z}$